CHURNQUEST

Navigating the Waves of Customer Retention in Telecommunications

A Customer Churn Analysis Case Study Using Microsoft Excel

Industry: Telecommunications
Business Context: Airtel
Domain: Customer Analytics and Retention
Tools: Microsoft Excel, Power Query, PivotTables, PivotCharts
Dataset: Telecom Customer Churn Dataset
Total Customer Records: 4,250

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1. Introduction

Customer retention is one of the most important challenges faced by telecommunications companies. In a highly competitive market, customers can switch service providers because of factors such as pricing, service quality, customer support experiences, or changing communication needs.

Customer churn refers to the loss of customers when they discontinue their relationship with a service provider. High churn rates can negatively affect revenue, profitability, and long-term business growth.

Understanding customer churn is therefore essential for telecommunications companies seeking to improve customer satisfaction and develop effective retention strategies.

This case study, titled **"ChurnQuest: Navigating the Waves of Customer Retention in Telecommunications,"** explores customer churn patterns using a telecommunications customer dataset.

The analysis is conducted from the perspective of a data analyst working with Airtel in a hypothetical business scenario. The objective is to examine customer characteristics, service subscriptions, calling behavior, customer support interactions, and geographic differences to identify factors associated with customer churn.

2. Business Background

Airtel operates in the competitive telecommunications industry, where maintaining customer loyalty is essential for sustainable growth.

The company offers services such as voice calling, international calling, and additional communication features.

Despite these offerings, some customers discontinue their services, creating challenges for customer retention and revenue stability.

The management team wants to better understand customer behavior and identify groups that may be more likely to discontinue their services.

To support this objective, the analytics team has been assigned a customer churn analysis project using historical customer information.

The findings will help management explore opportunities to improve service quality, optimize pricing strategies, strengthen customer relationships, and prioritize retention initiatives.

3. Problem Statement

Customer churn presents a significant business challenge because losing existing customers can reduce recurring revenue and increase the need for customer acquisition.

However, without analyzing customer data, it is difficult to identify which customer characteristics and behaviors are associated with churn.

The central business problem addressed in this case study is:

"How can a telecommunications company use customer data to identify churn patterns, recognize high-risk customer segments, and develop data-driven strategies to improve customer retention?"

The analysis aims to answer this question by examining multiple dimensions of customer behavior and service usage.

4. Project Objectives

The primary objective of this project is to analyze customer churn and generate actionable business insights using Microsoft Excel.

The specific objectives are:

1. Determine the overall customer churn rate.
2. Examine the relationship between customer tenure and churn.
3. Evaluate whether international plan subscriptions are associated with churn.
4. Analyze the relationship between voicemail plan subscriptions and customer retention.
5. Investigate whether frequent customer service calls are associated with higher churn.
6. Identify customer groups exhibiting multiple churn-related characteristics.
7. Compare calling patterns between churned and retained customers.
8. Examine international calling usage and its relationship with churn.
9. Compare calling charges between churned and retained customers.
10. Identify geographic differences in customer churn rates.
11. Develop customer segments based on service calls and subscription characteristics.
12. Recommend data-driven customer retention strategies.

5. Dataset Overview

The project uses a telecommunications customer churn dataset containing information about **4,250 customers**.

Each record represents an individual customer and includes details about their account, service subscriptions, communication usage, billing charges, and churn status.

5.1 Dataset Characteristics

Attribute	Description
Dataset Type	Telecommunications Customer Churn
Number of Records	4,250
Original Number of Columns	20
Data Format	CSV
Analysis Tool	Microsoft Excel
Data Preparation Tool	Power Query
Target Variable	Churn
Geographic Information	US States

5.2 Key Dataset Variables

Customer Information

- State: Geographic location of the customer.
- Account Length: Duration of the customer's account.
- Area Code: Telephone area code.

Service Subscriptions

- International Plan: Indicates whether the customer subscribes to an international calling plan.
- Voice Mail Plan: Indicates whether the customer subscribes to a voicemail service.

Calling Behavior

- Total Day Minutes: Total daytime calling minutes.
- Total Evening Minutes: Total evening calling minutes.
- Total Night Minutes: Total nighttime calling minutes.
- Total International Minutes: Total international calling minutes.

Billing Information

- Total Day Charge: Charges associated with daytime calls.
- Total Evening Charge: Charges associated with evening calls.
- Total Night Charge: Charges associated with nighttime calls.
- Total International Charge: Charges associated with international calls.

Customer Service

- Customer Service Calls: Number of customer service calls made by the customer.

Target Variable

- Churn: Indicates whether a customer discontinued their service.

6. Data Preparation and Transformation

Before performing the analysis, the dataset was imported into Microsoft Excel and prepared using Power Query.

The data preparation process involved reviewing data quality, verifying column types, and creating additional fields to support the analysis.

6.1 Data Quality Assessment

The dataset was examined for:

- Missing values
- Duplicate records
- Inconsistent data types
- Incorrect categorical values
- Data formatting issues

The initial data review identified no missing values or duplicate customer records.

6.2 Data Transformation

Additional calculated columns were created to simplify customer segmentation and churn analysis.

Churn Flag

A binary churn indicator was created:

- 1 = Customer churned
- 0 = Customer retained

This variable enables the calculation of churn counts and churn rates through Excel PivotTables.

Tenure Group

Customers were divided into account-length categories:

- 0–23 months
- 24–59 months
- 60–95 months
- 96–131 months
- 132+ months

These categories enable comparisons of churn across different account-tenure groups.

Customer Service Call Group

Customers were classified into two groups:

- 0–3 customer service calls
- 4 or more customer service calls

This segmentation helps evaluate whether frequent customer service interactions are associated with customer churn.

7. Analytical Methodology

The project follows a descriptive and exploratory data analysis approach.

Microsoft Excel was used to summarize customer information, calculate churn metrics, compare customer groups, and visualize important patterns.

7.1 Tools and Techniques

Microsoft Excel

Used for organizing customer information, calculating metrics, and presenting the analysis.

Power Query

Used for data preparation, transformation, and creation of calculated columns.

PivotTables

Used to aggregate customer records and calculate customer counts, churn counts, average usage, average charges, and churn rates.

PivotCharts

Used to visualize important relationships and communicate analytical findings.

Excel Dashboard

Used to summarize the most important customer churn metrics and business insights in a single view.

7.2 Key Performance Indicators

The following metrics were used throughout the analysis:

Total Customers

The total number of customer records in the dataset.

Churned Customers

The number of customers who discontinued their service.

Retained Customers

The number of customers who did not discontinue their service.

Customer Churn Rate

Calculated as:

Churn Rate (%) = (Number of Churned Customers / Total Customers) × 100

Average Usage

The average number of calling minutes for a particular customer group.

Average Charges

The average calling charges incurred by customers in a particular group.

Segment Churn Rate

The percentage of customers who churned within a defined customer segment.

8. Business Questions

The case study focuses on 12 analytical questions organized into three categories.

8.1 Basic Analysis

Question 1: What is the overall customer churn rate, and how many customers have churned versus remained with the company?

Question 2: How does account tenure differ between churned and retained customers, and how do churn rates vary across tenure groups?

Question 3: Is international plan subscription associated with differences in customer churn rates?

Question 4: How do churn rates differ between customers with and without a voicemail plan?

8.2 Intermediate Analysis

Question 5: Are customers who make four or more customer service calls more likely to churn than customers who make fewer calls?